



FREE GUIDE · 2026 EDITION

Building with Light Gauge Steel in Bali & Indonesia

Costs, timelines and the build process for steel-frame villas, townhouses and commercial buildings, explained in plain English by the team that makes the frames.

Infinity Living Options

Own factory in Banyuwangi, East Java
Building across Bali, Lombok and Indonesia

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What is light gauge steel?

Light gauge steel (LGS, called baja ringan in Indonesia) is a way of framing a building with thin, galvanised steel sections instead of a concrete column-and-beam frame or timber. Steel coil is roll-formed at room temperature into C-shaped studs, tracks and joists. Cold-forming work-hardens the steel, and load-bearing members meet ASTM standards with minimum yield strengths of 230 to 340 MPa depending on thickness.

The frame is designed on computer, made in a factory, then delivered to site as finished wall panels, floor cassettes and roof trusses that are fixed together. Your foundations, cladding, roofing, services and finishes then go on as with any building.



Floor cassettes from the ILO factory, ready for site.

THE FRAME IN THREE PARTS

Walls

C-shaped studs between top and bottom tracks, with X-pattern bracing against wind and earthquake loads.

Floors

C-section joists on C-section bearers, sized to the loads on each floor.

Roofs

Steel trusses for hip, gable or dutch gable roofs, for metal sheet or tiles, screwed directly to the walls.

Key facts

Material	Cold-formed galvanised steel with a zinc or aluminium-zinc coating against rust.
Strength	ASTM-compliant load-bearing members, 230 to 340 MPa minimum yield strength.
Lifespan	More than 50 years structural lifespan when properly installed and enclosed.
Building types	Villas, townhouses, resorts and commercial buildings. LGS can reach mid-rise; most Bali homes are 2 to 3 storeys.
Services	Pre-punched holes in the studs carry electrical, plumbing and air-conditioning lines without on-site drilling.

BUILT FOR THE TROPICS

Why steel frames suit Bali and Indonesia

Bali's humidity, termites and earthquakes are the three things that wear down a building here. A galvanised steel frame is designed to deal with all three.

01

Earthquake resistant

Steel frames are much lighter than concrete, so they attract much lower earthquake forces. Steel is ductile: it can flex and absorb the energy instead of cracking. Indonesia sits on the Pacific Ring of Fire.

02

Termite and rot proof

Steel is inorganic. Termites cannot eat it and it cannot rot, warp or grow mould, so there is no need for chemical termite treatments common with timber.

03

Built 2 to 3 times faster

Frames are made in our factory while your site and foundations are prepared, then assembled on site in days rather than weeks. Rental income or move-in starts sooner.

04

Will not burn

Steel is non-combustible. It does not ignite or add fuel to a fire, which improves safety and can help with insurance.

05

Stable in humidity

Steel does not swell, shrink or warp with moisture or temperature. Walls stay straight and doors and windows keep closing properly.

06

Recyclable, little waste

Steel is 100% recyclable. Frames are cut to size in the factory: one real ILO wall panel used 36.5 m of steel with just 153 mm of scrap (0.4%).

Design freedom

Steel's strength allows open-plan living, larger spans and cantilevers, which suits tropical villas that open the living areas to the garden and pool.

PLANNING YOUR BUDGET

What does it cost?

ILO prices a build by how far it goes. The rates below are the published planning ranges for villas in Bali, per square metre of floor area. Townhouse and commercial rates are usually lower per square metre.

BUILD STAGE	PLANNING RATE	WHAT'S INCLUDED
Frame & Roof	From about Rp 2.5 million/m ²	Engineered LGS frame, floor cassettes, roof structure and sheeting, on your foundations.
Lockup Shell	About Rp 4 million/m ²	Weather-tight lockup: frame, ILO envelope system, roofing, doors and windows.
Turnkey	Rp 7 to 11 million/m ² depending on finish	Complete build: frame, envelope, interiors, services, tiling, move-in ready. The top of the range covers premium finishes such as stone, timber cladding, upgraded sanitaryware and joinery.

QUICK SUM, NOT A QUOTE

150 m² turnkey villa at Rp 7 to 11 million/m²

About Rp 1.05 to 1.65 billion

Simple multiplication of the planning range. Your quote depends on your design, site and finishes.

Why the total is usually lower

Once the shorter build time, lower labour and minimal material waste are counted, total LGS project costs are typically 25 to 40% lower than traditional brick and concrete.

What moves the price

- Floor area and number of storeys
- Finish level, from standard to premium
- Site access and location
- Design: spans, cantilevers and roof shape

BUILDERS AND DEVELOPERS: FRAME KITS ONLY

Already have a builder or your own crew? Our Infinity Structural frame kits are supplied only, or supplied and erected, in Bali and Lombok. A single-storey kit is from about Rp 660,000 per m² of floor area ex-factory, based on an 11 kg/m² planning assumption. Two-storey, hybrid and installed packages are quoted per project. See www.ilo.net.au/structural.

All rates are indicative planning references from ILO's published 2026 rates. They exclude permits and site-specific works, and depend on project, scope and engineering. A written quotation follows a free consultation.

HEAD TO HEAD

Steel frame vs brick and concrete

	LIGHT GAUGE STEEL	TRADITIONAL BRICK / CONCRETE
Build speed	2 to 3 times faster	Slower, with curing time and weather delays
Total cost	Typically 25 to 40% lower overall	Higher overall cost
Earthquakes	Light and flexible; absorbs energy	Heavy and rigid; prone to cracking
Termites	Immune	Timber elements are vulnerable
Humidity	No rot, no warping	Risk of moisture damage
Fire	Non-combustible	Depends on materials
Quality	Factory precision	Varies with each site crew
End of life	100% recyclable	Mostly landfill

Common myths, answered

"Steel will rust."

Galvanised coatings protect the steel. Properly installed and enclosed, the frame has a structural lifespan of more than 50 years.

"Steel costs more."

Per kilo it can. Across the whole project, faster building, less waste and lower labour bring the total down.

"Steel houses are noisy."

Sound depends on how the wall is insulated and lined, not on the frame itself.

"Steel is only for small buildings."

LGS is used for load-bearing walls and mid-rise buildings, well beyond the 2 to 3 storeys common in Bali.

DESIGN TO KEYS

How an ILO build works, step by step

01

Consultation and planning

We look at your site, budget, brief and design ideas, check that steel suits the project and outline the scope.

02

Architectural and structural design

Our engineers produce CAD drawings designed for steel framing, Indonesian seismic requirements and the tropical climate.

03

Permits

We help with PBG building permits and local compliance, so you can focus on the project.

04

Factory manufacturing

Galvanised coil is roll-formed into studs, tracks and joists in our Banyuwangi factory, cut by computer-controlled machines.

05

Frame prefabrication

Parts are assembled into finished wall frames, floor cassettes and roof trusses indoors, away from the weather.

06

Transport

Frames are packed and shipped from Banyuwangi to your site in Bali, Lombok, Java or elsewhere in Indonesia.

07

On-site assembly

Our crews stand the wall panels, floors and trusses in days, while your site is ready and waiting.

08

Finishing and handover

Cladding, linings, electrical, plumbing and air-conditioning, then finishes, a final inspection and handover with full documentation.



The ILO factory in Banyuwangi, East Java.



Frames going up on an ILO site in Canggu.

WHY THE FACTORY MATTERS

Making the frame indoors removes weather delays, keeps every part to size and lets frames be built while the foundations are poured. That overlap is where most of the time saving comes from.

NEXT STEP

Get a price for your project

Send us what you have, even a sketch. The first consultation is free and we reply by email or WhatsApp.

What to send us

- Drawings, a concept plan or photos of a design you like
- Where the land is, and whether you have it yet
- Approximate floor area and number of storeys
- The stage you want: frame and roof, lockup shell or turnkey
- When you would like to start

VILLA OWNERS AND INVESTORS

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BUILDERS AND DEVELOPERS

Frame kits: supply or supply + erect

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Try the Build & Price tool

Pick your building type, stage and size for a live estimate at www.ilo.net.au/#configurator